Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-3

Number of questions: 15 **Total marks: 10 X 1= 15**

1. Which of the following is not a feature of Cyber-Physical Systems?

a. Reactive Computation
b. Concurrency
c. Real-Time Computation
d. Standalone non-networked Computation

Answer: (d)

Justification: Please refer to Slides 6 and 7 of Lecture 1 of Week 3

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2. In the context of CPS in Smart Grid, what is the full form of DAU?

a. Data Aggregation Unit
b. Data Analytic Unit
c. Distributed Augmented Universe
d. Dedicated Accounting Unit

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 1 of Week 3

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3. Which of the following is not a component of CPS's “5C Architecture” for Industry 4.0?

a. Connection
b. Conversion
c. Compatibility
d. Cognition

Answer: (c)

Justification: Please refer to Slide 12 of Lecture 1 of Week 3

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4. What are the key parts enabling collaboration productivity in Industry 4.0?

a. IT Proliferation, Single Source of Truth, Industrialization, Coordination
b. Financial Analysis, Market Research, Legal Compliance, Public Relations
c. Human Resources, Sales Targets, Customer Feedback, Advertising
d. Supply Chain Management, Quality Control, Cost Reduction, Branding

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 2 of Week 3

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5. What does 'PLM' stand for in the context of Industry 4.0?

a. Product Lifecycle Management
b. Public Liability Marketing
c. Professional Learning Modules
d. Personal Leadership Mastery

Answer: (a)

Justification: Please refer to Slide 9 of Lecture 2 of Week 3

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6. What does the Technique 'ABC' in PLM stand for?

a. Activity Based Costing
b. Advanced Business Computing
c. Analytical Business Concepts
d. Automated Banking Coordination

Answer: (a)

Justification: Please refer to Slide 21 of Lecture 2 of Week 3

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7. In which fields are AR and VR used in Industry 4.0?

a. Cooking and nutrition
b. Fashion design only
c. Machining, production, and education
d. Network Security

Answer: (c)

Justification: Please refer to Slides 2 of Lecture 3 of Week 3

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8. Which among the following is one type of categorization of AR?

a. Marker-attached AR
b. Markerless AR
c. Null AR
d. Marker-like AR

Answer: (b)

Justification: Please refer to Page 153 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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9. Who first coined the term "Augmented Reality"?

a. Thamos Caudell
b. Steve Jobs
c. Elon Musk
d. Mark Zuckerberg

Answer: (a)

Justification: Please refer to Slide 6 of Lecture 3 of Week 3

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10. What type of AR uses visual markers for outcomes?

a. Marker-based Augmented Reality
b. Markerless Augmented Reality
c. Projection Based Augmented Reality
d. Superimposition Based Augmented Reality

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 3 of Week 3

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11. What type of knowledge is used in AI techniques?

a. Only historical data
b. Only factual information
c. Procedural, declarative, and heuristic knowledge
d. Limited to scientific formulas

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 4 of Week 3

12. What is the use of AI in IIoT?

a. No use in IIoT
b. Only for data storage
c. Machine safety, efficient product lifecycle, efficient manufacturing processes, and others
d. Limited to basic calculations

Answer: (c)

Justification: Please refer to Slide 11 of Lecture 4 of Week 3

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13. Which of the following characterizes Big Data?

a. Limited to textual data only
b. Consists of structured and non-structured data
c. Only includes numerical data
d. Data that is always organized

Answer: (b)

Justification: Please refer to Slide 2 of Lecture 5 of Week 3

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14. What does a distributed file system in Big Data ensure?

a. Inconsistency of data
b. Unreliable data storage
c. Consistency, accessibility, and fault tolerance of data
d. Limited data access

Answer: (c)

Justification: Please refer to Slide 18 of Lecture 5 of Week 3

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15. Suppose you write a program which uses API to collect data continuously from LinkedIn and Facebook. You then wish to analyze such data. What type of big data source you are using?

a. Internal data source
b. External data source
c. Complimentary data source
d. All of the above

Answer: (b)

Justification: Please refer to Page 163 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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